

Program1: Build a docker container from a Dockerfile

step1: create directory also create app.py and Dockerfile inside the directory

step2: build the docker

```
isha@neow:~$ mkdir Program-1
cd Program-1
isha@neow:~/Program-1$ nano Dockerfile
isha@neow:~/Program-1$ nano app.py
isha@neow:~/Program-1$ nano requirements.txt
isha@neow:~/Program-1$ sudo docker build -t program-1 .
[+] Building 37.1s (10/10) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 268B                                0.0s
=> [internal] load metadata for docker.io/library/python:3.9-slim 15.4s
=> [internal] load .dockerignore                                   0.0s
=> => transferring context: 2B                                       0.0s
=> [internal] load build context                                   0.0s
=> => transferring context: 528B                                     0.0s
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:545badebace9a958b98d3e272f0f0d46c0a1a389ac77e24c33f2e7b548ce1b6b 13.9s
=> => resolve docker.io/library/python:3.9-slim@sha256:545badebace9a958b98d3e272f0f0d46c0a1a389ac77e24c33f2e7b548ce1b6b 0.0s
=> => sha256:725dd64d2f62808e0aee4d5d916f4c208275716e74cc700ed68aa17078411ca9 5.30kB / 5.30kB 0.0s
=> => sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 29.78MB / 29.78MB 11.5s
=> => sha256:c57466e36deb99f355eab1131bc21d799709189ecdd7fb41cf4c17d5f53724e2 1.29MB / 1.29MB 3.4s
=> => sha256:6f010d693edcb2d965ff3c2a741fc92cdd69746aa616f746e7813896fdd443b 13.87MB / 13.87MB 8.8s
=> => sha256:545badebace9a958b98d3e272f0f0d46c0a1a389ac77e24c33f2e7b548ce1b6b 10.36kB / 10.36kB 0.0s
=> => sha256:dc9f486a31b5c0642ed5606b41c820fae66e6d91003c32e4f3d1624b6e7425cd 1.74kB / 1.74kB 0.0s
=> => sha256:21ed5e7e522581efdab14b4fde1d7d493901b4d25736b947520695c7911da66b 250B / 250B 4.5s
=> => extracting sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 1.1s
=> => extracting sha256:c57466e36deb99f355eab1131bc21d799709189ecdd7fb41cf4c17d5f53724e2 0.1s
=> => extracting sha256:6f010d693edcb2d965ff3c2a741fc92cdd69746aa616f746e7813896fdd443b 0.8s
=> => extracting sha256:21ed5e7e522581efdab14b4fde1d7d493901b4d25736b947520695c7911da66b 0.0s
=> [2/5] WORKDIR /app                                             0.1s
=> [3/5] COPY requirements.txt .                                   0.0s
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt       7.1s
=> [5/5] COPY . .                                                 0.0s
=> exporting to image                                             0.3s
=> => exporting layers                                             0.2s
=> => writing image sha256:089fb29f2b5400102e6a9d656dbb3f7400d5759c3936429ac04018a3e48dce22 0.0s
=> => naming to docker.io/library/program-1                       0.0s
isha@neow:~/Program-1$
```

GNU nano 7.2	Dockerfile
FROM python:3.9-slim	
WORKDIR /app	
COPY requirements.txt .	
RUN pip install --no-cache-dir -r requirements.txt	
COPY . .	
ENV FLASK_APP=app.py	
ENV FLASK_RUN_HOST=0.0.0	
ENV FLASK_RUN_PORT=5000	
EXPOSE 5000	
CMD ["flask", "run"]	

step3: connect to port to see the output

This is app.py file

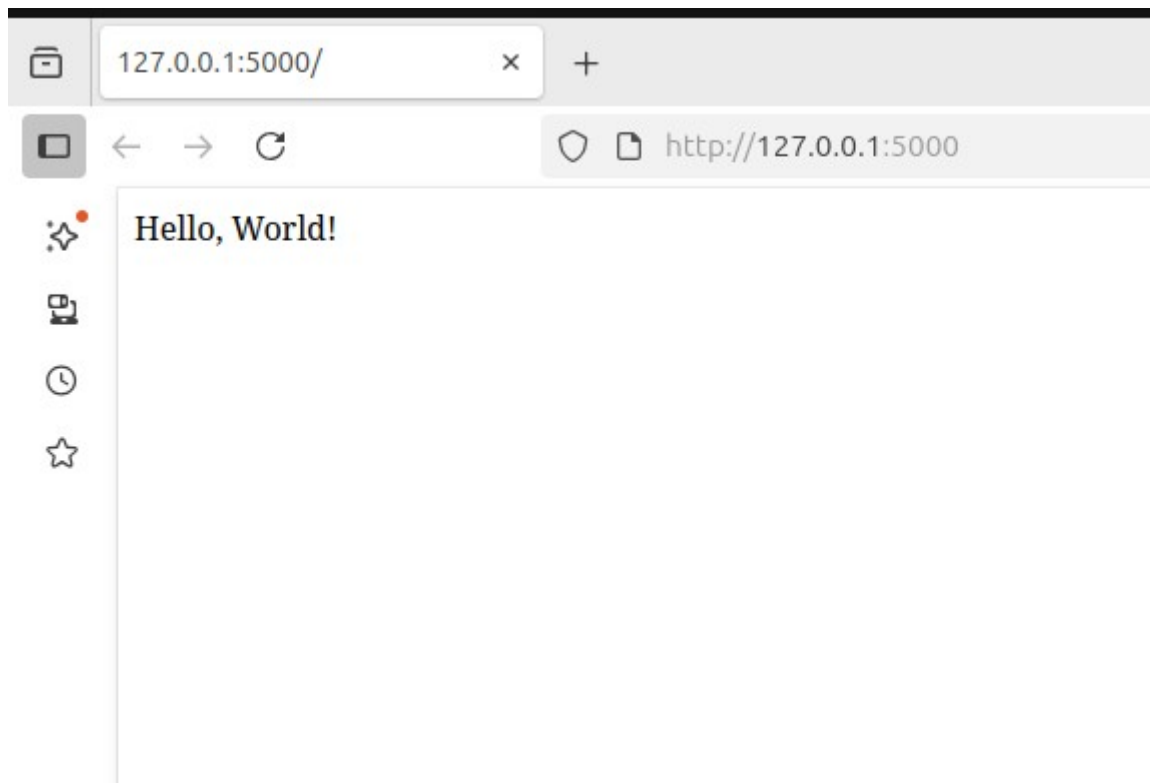
```
GNU nano 7.2 app.py
from flask import Flask

app = Flask(__name__)

@app.route("/")
def hello():
    return "Hello, Docker!"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

step4: This is the output



This is Dockerfile(no extension)

```
GNU nano 7.2 Dockerfile
FROM python:3.9-slim

WORKDIR /app

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY . .

ENV FLASK_APP=app.py
ENV FLASK_RUN_HOST=0.0.0.0
ENV FLASK_RUN_PORT=5000

EXPOSE 5000

CMD ["flask", "run"]
```